# Sean Nian

https://seannian.github.io/ | snian2@illinois.edu | linkedin.com/in/seannian | 1(408)-XXX-XXXX

## Education

**University of Illinois Urbana-Champaign** — Champaign, IL
Doctor of Philosophy in Computer Science — Aug. 2025 – Aug. 2030
*Advisor*: Prof. Fan Lai
*Research Interests*: Machine Learning Systems, Systems for GenAI, Systems for Agents

**San Jose State University** — San Jose, CA
Bachelor of Science in Computer Science with Honors — Aug. 2021 – Dec. 2024

## Publications

**S. Nian**, J. Fang, Q. Feng, Z. Wu, and F. Lai. *CacheFlow: Accelerating LLM Serving via 3D Parallel KV Cache Restoration.* [Conference Paper]. In Submission.

**S. Nian** and G. Ishigaki. *Autoscaling in Knative for Serverless Computing Using Deep Reinforcement Learning.* [Conference Paper]. IEEE Annual Congress on Artificial Intelligence of Things (IEEE AIoT), Osaka, Japan, 2025, pp. 630-634. DOI: 10.1109/AIoT66900.2025.00094.

K. Hollingsworth, **S. Nian**, A. Gutierrez, and A. Padmanabhan. *An Analysis of Network Overhead in Distributed TinyML.* [Workshop Paper]. IEEE/ACM Symposium on Edge Computing (SEC), Rome, Italy, 2024, pp. 449-455. DOI: 10.1109/SEC62691.2024.00051.

## Presentations

*An Analysis of Network Overhead in Distributed TinyML.* [Sole Presenter]. INTERACT Workshop, IEEE/ACM Symposium on Edge Computing (SEC), Rome, Italy, 2024.

*An Analysis of Network Overhead in Distributed TinyML.* [Poster, Co-presented]. Harvey Mudd College CS Summer Research Multi-Institutional Seminar, 2024.

## Other

**S. Nian**, A. Huang, and B. Reed. *Building a Mastodon Compatible Java Server for ActivityPub.* [Preprint]. arXiv:2412.09011 [cs.SI], 2024. DOI: 10.48550/arXiv.2412.09011.

S. Qi, X. I. Quan, T. Park, and B. Makani. *The Role of Artificial Intelligence in Enhancing Scholarly Research – AI Tools Evaluation.* [Acknowledged Research Assistance: **S. Nian**]. Quarterly Review of Business Disciplines, vol. 11, no. 3/4, Feb. 2025, pp. 90-114.

## Research

**AgentCompiler: A Fault-Tolerant Runtime for Agentic Systems** — Dec. 2025 – Present
University of Illinois Urbana-Champaign, advised by Prof. Fan Lai

- Designing a fault-tolerant runtime that utilizes speculative execution and lightweight checkpointing to mitigate cascading failures, significantly reducing costs for long-running agentic workflows

**CacheFlow: Efficient 3D-Parallel KV Cache Restoration** — Aug. 2025 – Mar. 2026
University of Illinois Urbana-Champaign, advised by Prof. Fan Lai

- Designed a 3D-parallel KV cache restoration framework spanning token-wise, layer-wise, and multi-GPU dimensions, enabling fine-grained overlap of recomputation and I/O during LLM prefill
- Developed a batch-aware two-pointer scheduler that coordinates compute and I/O across concurrent requests, prioritizing I/O for longer prefixes to avoid costly quadratic recomputation
- Implemented CacheFlow atop vLLM and LMCache, achieving 10%–62% TTFT reduction over state-of-the-art baselines across Qwen3-8B, Llama-3.1-8B, and Qwen3-30B-A3B

**ML Systems for Serverless Computing** — Aug. 2024 – April 2025
San Jose State University, advised by Prof. Genya Ishigaki

- Developed a Deep Reinforcement Learning (DRL) agent using PPO to perform simultaneous horizontal (pod count) and vertical (CPU/memory) scaling in Knative
- Achieved superior performance over native autoscalers, reducing memory consumption by up to 44% while increasing request throughput by 18.6% in high-concurrency scenarios
- Engineered a custom OpenAI Gym environment wrapping a live Kubernetes cluster to train the agent and programmatically apply scaling decisions by modifying Knative manifests in real-time

**Optimizing TinyML Systems** — May 2024 – July 2024
Harvey Mudd College (NSF Funded REU), advised by Prof. Arthi Padmanabhan

- Engineered a distributed TinyML inference system using C++ on ESP32-S3 microcontrollers by partitioning a quantized TensorFlow MobileNet model and distributing intermediate computations over Bluetooth Low Energy (BLE)

- Optimized the BLE network communication protocol for enhanced fault-tolerance and energy efficiency, analyzing network overhead to reduce the distributed inference failure rate to 0%
- Authored a 6-page paper on the network overhead and challenges of distributed TinyML, accepted for publication and presentation at the INTERACT workshop at ACM/IEEE SEC 2024

**Decentralized Social Media Networks** Jun. 2023 – May 2024
San Jose State University, advised by Prof. Ben Reed

- Explored decentralized social media networks by analyzing the ActivityPub Protocol's architecture
- Engineered moth, an open-source ActivityPub server in Java, by implementing RESTful API endpoints with JSON payloads using Spring WebFlux to ensure robust decentralized communication
- Reverse-engineered Mastodon's client-server API by analyzing network traffic with Wireshark, enabling full compatibility between our moth server and the official Mastodon web frontend

## Industry Experience

**Google Research** Incoming May 2026
PhD Student Researcher Seattle, WA

- Incoming SystemsResearch@Google

**PACCAR Silicon Valley Innovation Center** May 2025 – Aug. 2025
R&D Software Engineer Intern Sunnyvale, CA

- Architected an edge-to-cloud data pipeline for an AI-powered dashcam system in Kotlin using Android Studio with low-level libIPC components, optimizing for resource-constrained hardware
- Engineered a vehicle data processing module by creating custom DBC files compliant with SAE J1939 standards; decoding 60+ CAN messages, mapping over 600 signals and 20,000 PGN/SPNs
- Implemented an edge management system to prevent on-device storage exhaustion, implementing a RAM and ROM monitoring system to automatically offload encrypted files to the cloud
- Enabled fleet-wide data visualization and analysis, creating a scalable RESTful API using Node.js and Express.js to ingest high-volume telemetry into the PACCAR data lake via MongoDB schemas
- Presented the end-to-end system at the PACCAR Innovation Center's Q3 review and briefed leadership at PACCAR Global Connected Services, demonstrating production viability and leading to plans for continued development

## Awards

2025 NSF Graduate Research Fellow (NSF GRFP)
2024 NSF Student Travel Grant IEEE/ACM SEC – $1,700
2021 SJSU Gail Fullerton Endowment – $1,000
2021 SJSU XILINX Scholarship – $2,500

## References

**Prof. Fan Lai**
Assistant Professor of Computer Science at the University of Illinois Urbana-Champaign,
Email: fanlai@illinois.edu.

**Prof. Genya Ishigaki**
Assistant Professor of Computer Science at San Jose State University, Email: genya.ishigaki@sjsu.edu.

**Prof. Arthi Padmanabhan**
Assistant Professor of Computer Science at Harvey Mudd College, Email: arpadmanabhan@g.hmc.edu.

**Prof. Ben Reed**
Associate Professor of Computer Engineering at San Jose State University, Email: ben.reed@sjsu.edu.